



Presented by members of Dowsing:
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Project Need

Powder dispenser automatically measures the water you want and calculates the amount of powder (matcha) to dispense. This increases convenience and eliminates the need to manually measure

Overview

Other Examples

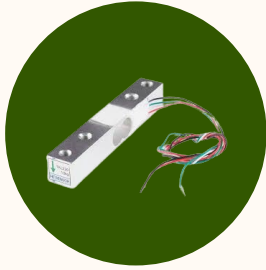
Most automatic powder dispensers like this are industrial-style, not suitable for household use



Problems Solved

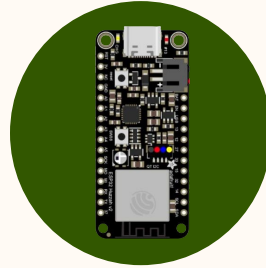
Most coffee-makers/dispensers require user input, but this weighs the cup directly and dispenses the optimal amount without having to be prompted. This saves time and ensures peak quality every time.

Sensing, Data Transmission, Actuation



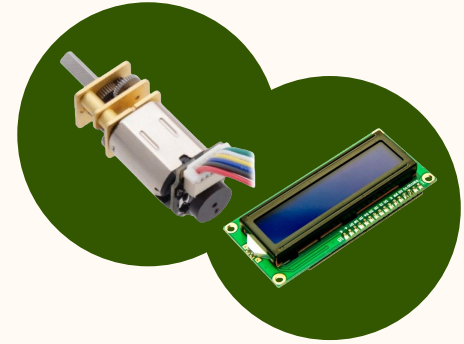
Sensing: Load Cell

Load cell (zeroed with the weight of the cup) measures the weight of the water



Data Transmission: ESP Now

ESPNow sends weight data from load cell to tell micromotor how much to turn



Actuation: Micromotor & LCD

Micromotor uses PWM signal to turn for a one interval of to dispense matcha in intervals, then LCD displays quantities measured & dispensed

Challenge/Solutions

Challenges:

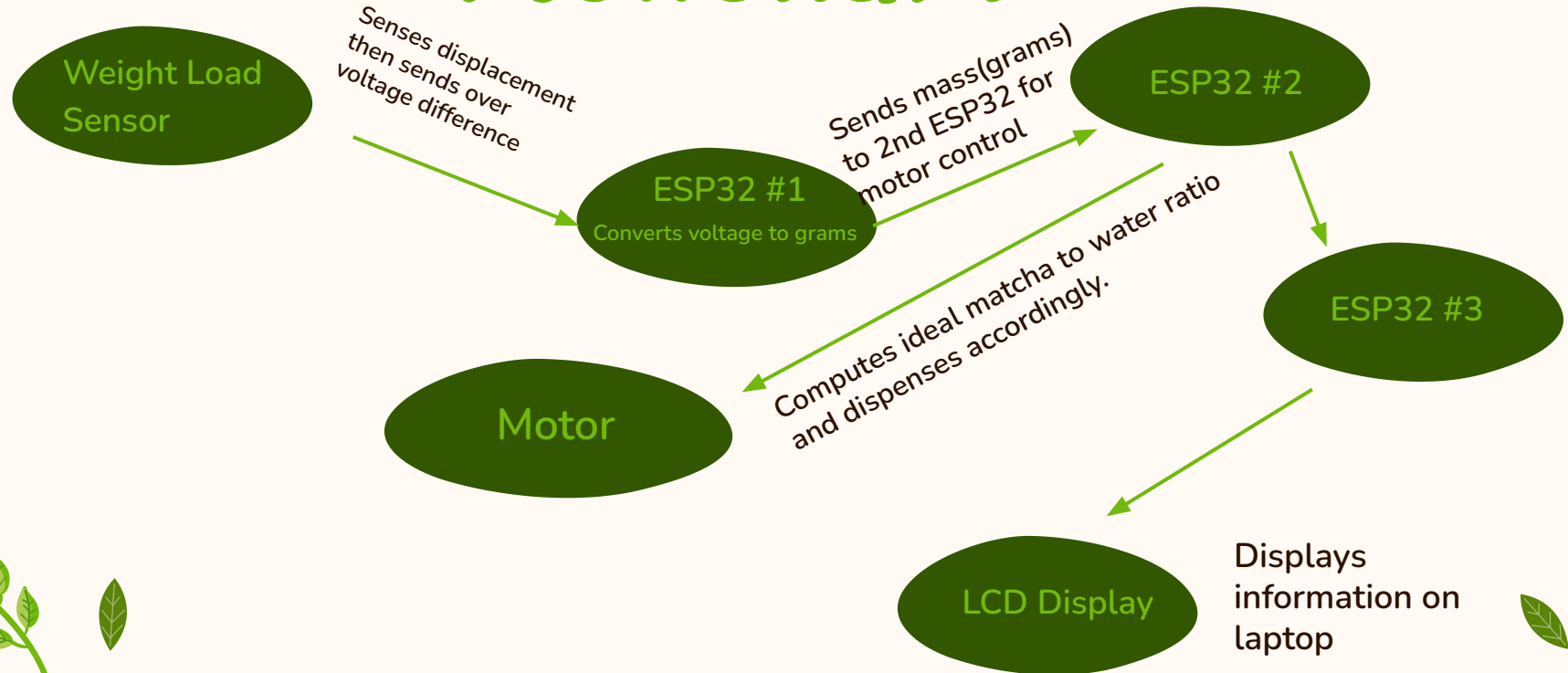
- Motor torque not enough
- Microgear motor does not have enough power
- No way to know how much water/matcha was involved

Solution:

- Increased tolerance between wheel and housing
- Utilized an external 9v battery source.
- Incorporated an LCD screen to output numbers.



Flowchart





LIVE DEMO: VIDEO ON

<https://drive.google.com/file/d/1sWx95r4op364dgnDKFqRugCXP4zL0sli/view?ts=69fba1f8>



